

SIH 2026 — Elaborative Problem Statements & Proposed Solutions

Selected codes: 003, 034, 090, 117, 140, 152

Basis: the supplied SIH 2026 HTML source. Source-derived requirements are separated from the proposed engineering solution. The proposed architecture, AI components and demo plans are recommendations for a hackathon prototype.

Special note on PS 034: its source explicitly asks for rule-based Legal Metrology compliance checking. AI/OCR is therefore proposed as an assistive extraction layer, not as a replacement for deterministic compliance rules.

PS 003 — AI-Based Cognitive Gaming and Memory Assistance Platform for Elderly Dementia Patients in North Eastern Region (NER)

SIH ID	SIH26003
Organization	Ministry of Development of North Eastern Region (MDoNER)
Theme	Space Technology
Category	Software

1. Problem Statement — Elaborated

The PS addresses dementia and memory loss among elderly people in the North Eastern Region, where access to specialized neurological care, cognitive therapy and long-term support is limited. It asks for an accessible AI-powered cognitive gaming and memory-assistance platform supporting memory improvement, attention, routine recall, pattern/object recognition, adaptive difficulty, multilingual/voice interaction, reminders, caregiver monitoring, offline use and an elderly-friendly mobile/tablet interface.

2. Official Expected Solution — Interpreted

A user-friendly AI-enabled cognitive assistance platform with adaptive gaming and memory-training modules, multilingual voice interaction, cognitive performance tracking, caregiver monitoring and alerts, offline synchronization, secure data management and accessible UI.

3. Proposed Solution

Build an adaptive cognitive-care platform in which each user gets a continuously updated performance profile. Cognitive games generate signals such as accuracy, reaction time, repetition errors and consistency. A personalization engine selects suitable activities and difficulty. Voice interaction handles simple commands/reminders, while a caregiver dashboard presents longitudinal trends. Position the system as assistance/monitoring, not autonomous diagnosis.

4. AI / ML Components

- Adaptive difficulty using a safe ML/rule hybrid or contextual recommendation policy.
- Performance modelling from accuracy, reaction time, completion rate and longitudinal trends.
- Speech-to-text/text-to-speech and multilingual intent handling.
- Personalized game and routine recommendations.
- Explainable caregiver alerts showing which performance indicators changed.

5. Proposed System Architecture

- Mobile/PWA client
- Python FastAPI/Flask backend
- PostgreSQL
- AI personalization service
- Speech/voice service
- Offline-first cache + sync
- Caregiver dashboard
- Encrypted records and audit logs

6. Suggested End-to-End Workflow / Demo

- Run several cognitive games.
- Show performance features being generated.
- Demonstrate AI-selected next difficulty.
- Use multilingual voice interaction.

- Show caregiver trend/alert dashboard.
- Demonstrate offline session followed by synchronization.

7. Hackathon Implementation Notes

Prioritize a demonstrable vertical slice over a large production platform: one complete user journey, one working AI model/service, clear input/output evidence, and a dashboard that makes the AI decision explainable. Use synthetic/public/demo data where the PS does not provide operational data.

8. Overall Assessment

Strong AI + health-tech opportunity; differentiation comes from adaptive personalization and voice-first accessibility.

PS 034 — Software System to check compliance of Packaged Commodities under Legal Metrology (Packaged Commodities) Rules, 2011 by scanning products, images and labels.

SIH ID	SIH26034
Organization	Ministry of Consumer Affairs, Food & Public Distribution
Theme	Agriculture, FoodTech & Rural Development
Category	Software

1. Problem Statement — Elaborated

The PS targets manual inspection of packaged products for mandatory declarations such as manufacturer/packer/importer details, net quantity, MRP, date information and consumer-care details. It asks for image/product scanning, declaration detection, correctness/completeness/placement checks, readability and font-size checks, violation reports, inspection history and enforcement dashboards.

2. Official Expected Solution — Interpreted

The source specifies a web/mobile application, automated extraction and validation, rule-based compliance checking, digital reports, dashboards, search/history, role-based access and PDF/editable exports.

3. Proposed Solution

Build a computer-vision + OCR compliance assistant. A package photograph is converted into structured fields; a deterministic rules engine then checks those fields against the applicable Legal Metrology requirements. AI should provide extraction and evidence, while the final compliance result remains traceable to explicit rules. This is important because the source explicitly asks for rule-based compliance checking rather than making AI mandatory.

4. AI / ML Components

- Image preprocessing and label/text-region detection.
- OCR and layout understanding for MRP, quantity, dates and manufacturer details.
- NLP normalization of units and equivalent wording.
- Vision-based readability/character-size estimation where image quality permits.
- Confidence scoring and human-review queue for ambiguous cases.

5. Proposed System Architecture

- Web/mobile upload
- Vision preprocessing + OCR
- NLP field extraction
- Deterministic compliance rules engine
- Evidence image crops + confidence
- Compliance report generator
- Official-review dashboard

6. Suggested End-to-End Workflow / Demo

- Upload package image.
- Highlight extracted declarations.
- Show structured fields.
- Run deterministic rules.
- Display missing/non-compliant fields with evidence.
- Generate compliance report.

7. Hackathon Implementation Notes

Prioritize a demonstrable vertical slice over a large production platform: one complete user journey, one working AI model/service, clear input/output evidence, and a dashboard that makes the AI decision explainable. Use synthetic/public/demo data where the PS does not provide operational data.

8. Overall Assessment

Strong computer-vision/OCR opportunity, but AI is not mandatory in the source; differentiation comes from evidence-backed extraction plus deterministic rules.

PS 090 — AI-Driven Market Linkage and Smart Cataloging Mobile Application for Marginalized Artisans

SIH ID	SIH26090
Organization	Ministry of Social Justice and Empowerment (MoSJE)
Theme	Miscellaneous
Category	Software

1. Problem Statement — Elaborated

Marginalized artisans and weavers often lack continuous digital-market access because of low digital literacy, language barriers and difficulty creating professional photographs, descriptions and prices. The PS asks for an AI-driven mobile application acting as a virtual business manager to digitize inventory, optimize listings and connect artisans with B2B buyers or government marketplaces.

2. Official Expected Solution — Interpreted

An AI-powered cross-platform mobile app with AI image enhancement, multilingual NLP cataloging from regional-language voice notes and an ML-based dynamic pricing assistant.

3. Proposed Solution

Create an artisan-first mobile commerce assistant: photograph a product, describe it by voice, and receive a ready-to-publish listing. AI cleans the image, extracts product attributes, translates/transcribes the voice note, generates professional listing text and estimates a price range from market/raw-material signals. A marketplace layer exposes the products to buyers.

4. AI / ML Components

- Computer-vision background removal, lighting correction and framing.
- Multilingual speech recognition and translation.
- LLM/NLP catalog generation: title, description, attributes and tags.
- Pricing recommendation from product attributes, market signals and raw-material costs.
- Similarity search for comparable products.

5. Proposed System Architecture

- Flutter/React Native
- Vision/image-enhancement service
- Speech/NLP service
- Product database
- Pricing/recommendation engine
- Marketplace/API layer
- Admin and artisan analytics

6. Suggested End-to-End Workflow / Demo

- Photograph a product.
- Show automatic image cleanup.
- Record a regional-language voice note.
- Generate Hindi/English listing.
- Show price range and factors.
- Publish to demo marketplace.

7. Hackathon Implementation Notes

Prioritize a demonstrable vertical slice over a large production platform: one complete user journey, one working AI model/service, clear input/output evidence, and a dashboard that makes the AI decision explainable. Use synthetic/public/demo data where the PS does not provide operational data.

8. Overall Assessment

Very strong practical AI opportunity; voice-to-catalog + image enhancement + pricing creates a compelling end-to-end demo.

PS 117 — Sovereign On-Premise Agentic AI Workbench using Open-Weight Multimodal LLMs for Confidential Industrial Work

SIH ID	SIH26117
Organization	Mangalore Refinery and Petrochemicals Limited (MRPL)
Theme	Smart Automation
Category	Software

1. Problem Statement — Elaborated

Industrial and government organizations handle confidential engineering drawings, P&IDs, financials, vendor negotiations, internal correspondence and sensitive operational data that cannot be sent to public cloud AI services. The PS asks for a self-hosted, air-gapped workbench on an organizational GPU server, multiple open-weight models, automatic model selection, agentic multi-step work, local tools, multimodal document understanding and local knowledge grounding with no external calls.

2. Official Expected Solution — Interpreted

A working local deployment demonstrating model auto-selection across task types, an end-to-end agentic task, sandboxed coding, multimodal understanding and visible proof that no external calls occur.

3. Proposed Solution

Build a local AI operating layer rather than a single chatbot. A model router selects an appropriate open-weight model; an agent planner decomposes tasks and calls permissioned local tools; local RAG grounds responses in manuals/SOPs; a sandbox verifies code/calculations; and network controls prove data never leaves the environment.

4. AI / ML Components

- Multi-model routing by task, context, latency and GPU capacity.
- Agent planner with controlled local file, code, spreadsheet and search tools.
- Multimodal understanding of scanned PDFs, drawings and images.
- Local RAG over manuals/SOPs with source citations.
- Sandboxed output validation and audit logging.

5. Proposed System Architecture

- On-prem GPU server
- Open-weight LLM/vision models
- Model router
- Agent orchestration
- Local vector DB/RAG
- Tool sandbox
- Document/PPT/Excel generation
- Egress controls + audit logs

6. Suggested End-to-End Workflow / Demo

- Load two local models.
- Show router choosing a model.
- Read a scanned inspection report.
- Retrieve local SOP evidence.
- Draft a Word approval note.
- Run a coding task in sandbox.
- Display network logs proving no external calls.

7. Hackathon Implementation Notes

Prioritize a demonstrable vertical slice over a large production platform: one complete user journey, one working AI model/service, clear input/output evidence, and a dashboard that makes the AI decision explainable. Use synthetic/public/demo data where the PS does not provide operational data.

8. Overall Assessment

Extremely strong AI engineering opportunity; the sovereign/local agentic architecture is the key differentiator.

PS 140 — AI-Based Interactive Quantum Algorithm Learning Platform

SIH ID	SIH26140
Organization	Egreen Quanta
Theme	Smart Education
Category	Software

1. Problem Statement — Elaborated

Quantum computing education is difficult because qubits, superposition, entanglement and quantum algorithms are abstract. Existing resources are often static/theoretical and access to quantum hardware is limited. The PS asks for an AI-powered interactive web platform for learning, designing, simulating and visualizing quantum algorithms, including circuit building, multiple simulators, AI tutoring, debugging, optimization suggestions, personalized learning, assessments and analytics.

2. Official Expected Solution — Interpreted

A comprehensive AI-based interactive quantum learning platform integrating structured content, circuit design, simulation, visualization, AI tutoring, assessment and progress analytics.

3. Proposed Solution

Build a browser-based quantum learning IDE. Students construct a circuit visually or in code, execute it in a simulator, inspect results and ask an AI tutor for explanations. The tutor should receive the actual circuit and curriculum context, so explanations are circuit-aware rather than generic chatbot responses.

4. AI / ML Components

- Curriculum-grounded AI tutor and Socratic hints.
- Circuit-aware debugging using gates, qubits and measurement results.
- Quantum SDK code generation/repair.
- Personalized learning path from assessment mistakes.
- Circuit optimization suggestions with explanations.

5. Proposed System Architecture

- React/Next.js web IDE
- Circuit editor
- Qiskit Aer/PennyLane/Cirq simulator layer
- Python backend
- LLM tutor + curriculum RAG
- Assessment database
- Instructor dashboard

6. Suggested End-to-End Workflow / Demo

- Open a lesson.
- Build a circuit with drag-and-drop.
- Run and visualize results.
- Introduce an error and ask AI to debug.
- Generate SDK code.
- Show personalized next lesson.

7. Hackathon Implementation Notes

Prioritize a demonstrable vertical slice over a large production platform: one complete user journey, one working AI model/service, clear input/output evidence, and a dashboard that makes the AI decision explainable. Use synthetic/public/demo data where the PS does not provide operational data.

8. Overall Assessment

Strong AI education opportunity with a distinctive quantum domain; a circuit-aware tutor is more defensible than a generic chatbot.

PS 152 — Social Media Analytics

SIH ID	SIH26152
Organization	National Technical Research Organisation (NTRO)
Theme	Miscellaneous
Category	Software

1. Problem Statement — Elaborated

The PS views social media as an ecosystem of sentiment, demographics, topics and influence networks. It asks for an AI-driven analytics framework that processes raw platform data to infer sentiment, map aggregate demographics, detect trends and analyze relationships. X and Telegram are listed as essentials, with Instagram/Facebook desirable and Reddit/YouTube as additions.

2. Official Expected Solution — Interpreted

An AI framework covering continuous data collection and timeline management, multidimensional sentiment inference, anonymized demographic profiling, real-time trend/topic detection and link/network analysis to identify influential nodes and visualize information or sentiment spread.

3. Proposed Solution

Build a time-aware social intelligence dashboard around a unified event schema. Ingest permitted data into a timestamped store, run NLP for sentiment/topic/entity analysis, maintain an interaction graph and combine temporal NLP signals with graph analytics to show emerging narratives and influential nodes. Use aggregate/anonymized demographics and explicit privacy controls.

4. AI / ML Components

- Multilingual sentiment/emotion and stance classification.
- Topic discovery, trend ranking and temporal forecasting.
- Entity/narrative tracking over time.
- Graph analytics for communities and influence.
- Anomaly detection for sudden sentiment/topic/network shifts.

5. Proposed System Architecture

- Platform ingestion adapters
- Timestamped event store
- NLP/embedding pipeline
- Vector database
- Temporal trend engine
- Graph analytics
- Timeline/network dashboard
- Privacy/anonymization layer

6. Suggested End-to-End Workflow / Demo

- Load sample X/Telegram data.
- Show historical timeline.
- Run sentiment/topic analysis.
- Detect and rank an emerging topic.
- Display influence/community graph.
- Replay timeline to visualize propagation.

7. Hackathon Implementation Notes

Prioritize a demonstrable vertical slice over a large production platform: one complete user journey, one working AI model/service, clear input/output evidence, and a dashboard that makes the AI decision explainable. Use synthetic/public/demo data where the PS does not provide operational data.

8. Overall Assessment

Very strong NLP + graph + temporal analytics opportunity; the combined narrative/influence view is the main differentiator.